

Pultec 3 Band EQ Documentation

Copyright © 2011 Ian Thompson-Bell

Introduction

This document aims to encapsulate all the design and build documentation for the **Pultec 3 band EQ PCBs**. This and the Helios 69 Tube EQ are built on identically sized PCBs (125 by 73mm) each with 3 PCB mounting Grayhill switches in identical positions to simplify front panel mounting.

Pultec 3 Band EQ

The Pultec 3 band PCB includes all the circuits found on the poor man's EQP1A project (PMEQP1A) and the mid boost/cut extension. It also includes space for an additional inductor (VTB9042) so that the hi boost EQ can be built as the original Pultec EQP1A. The frequencies employed are the same as in the PMEQP1A but you can of course change these by changing component values.

Pultec Hi Boost/Cut

The optimum component values and corresponding PCB reference for the hi boost/cut are shown in the table below. There are spaces on the PCB for up to two capacitors per boost/cut frequency.

Cpoorboost and Ccut are the capacitor values to use if building the original poor man's Pultec . If you do this then you also need to short together all the pins where the new inductor (LhiBoost) would have been, short out the pins of the connections to the HiQ pot and change the Qmax resistor to 4K7.

To build an inductive based hi boost just like the original EQP1A then you should use the Cboost capacitor values and fit the VTB9042 inductor. The original Pultec EQP1A did not have a Qmax resistor so the maximum Q was determined by circuit resistances and the dc resistance of the inductors themselves. Modern inductors tend to have lower dc resistance due to improved magnetic materials, so to obtain a maximum Q close to that of the original we add a series resistor, Qmax,. This has a nominal value of 470 ohms which has been found in tests to give a maximum Q similar to that indicated by the published curves of the original Pultec EQP1A.

Frequency	Lboost	Cboost	Cpoorboost	CboostRef	Ccut	CcutRef
3K	0.6H	4n7	15n + 1n	C7,C7A	47n +33n	C1,C1A
4K	0.3H	4n7 + 470pF	10n + 2n2	C8,C8A	47n + 10n	C2,C2A
5K	0.3H	3n3	10n	C9,C9A	47n	C3,C3A
8K	0.2H	1n + 1n	4n7 + 1n	C10,C10A	22n +10n	C4,C4A
10K	0.1H	1n + 1n5	4n7	C11,C11A	22n + 2n2	C5,C5A
16K	0.1H	1n	1n + 1n5	C12,C12A	15n	C6,C6A

Pultec Mid Boost/Cut

This version of the Pultec mid boost/cut implements the same 11 mid frequencies as the Pultec MEQ5 and uses a VTB9050 inductor. Optimum component values are shown in the table below.

Frequency	L	C
200	2H	330n
300	2H	150n
500	2H	47n
700	2H	$22n + 3n3$
1K	1H	$22n + 3n3$
1K5	1H	12n
2K	0.45H	$12n + 2n2$
3K	0.45H	$4n7 + 1n5$
4K	0.22H	$4n7 + 2n2$
5K	0.22H	4n7
7K	0.1H	$2n2 + 1n$

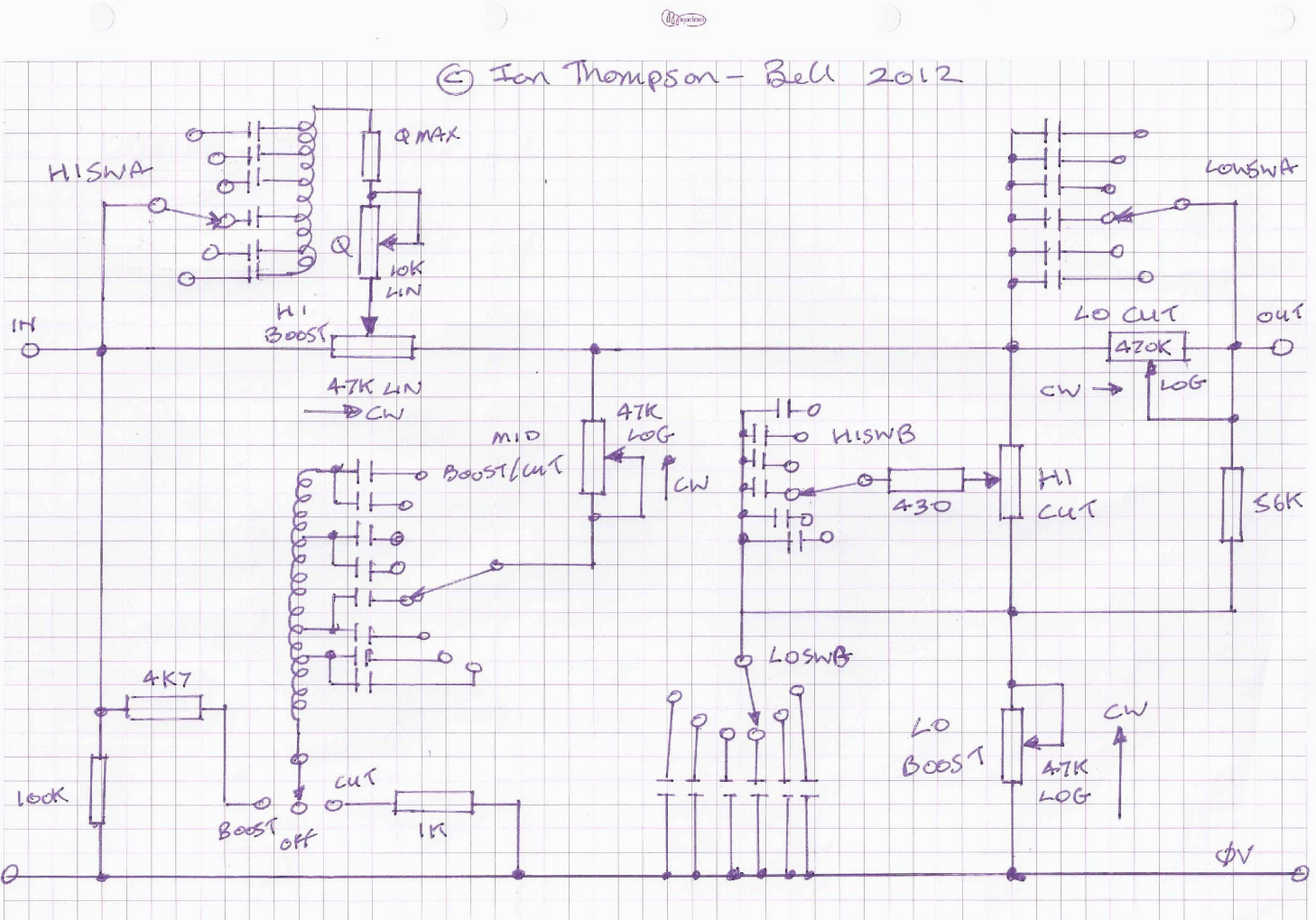
Pultec Lo Boost/Cut

The lo boost/cut frequencies are the same as in the PMEQP1A. Component values are shown in the table below. Single components are used for all but one cut frequency.

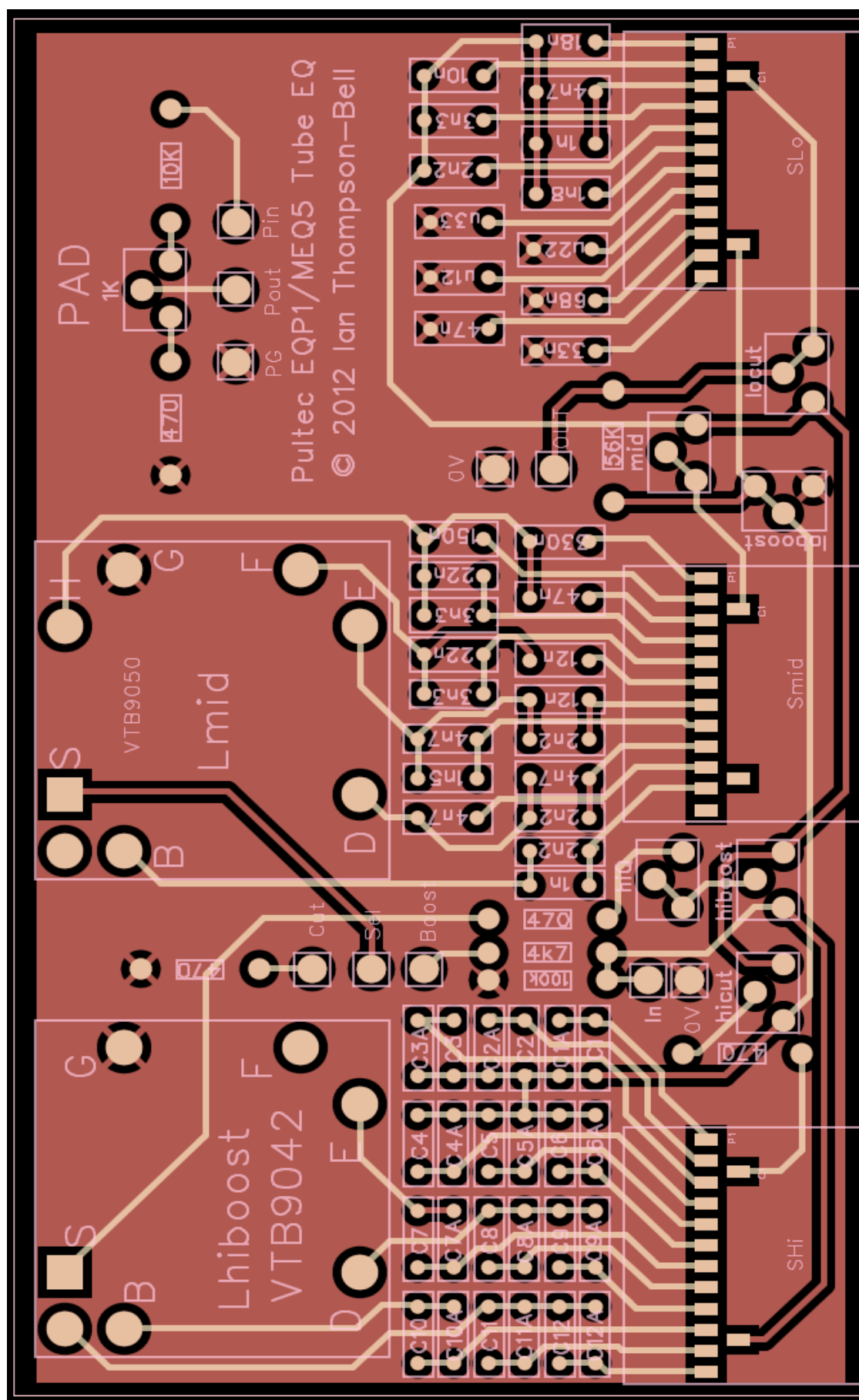
Frequency	Cboost	Ccut
20	330n	18n
30	220n	10n
60	120n	4n7 + 1n
100	68n	3n3
150	47n	2n2
200	33n	1n8

The Pultec 3 band schematic is shown below. Note, the output should be loaded with not less than 470K. As mentioned earlier, to build the hi boost as the original PMEQP1A you just need to substitute the correct capacitor values, short together all the pins where the hi boost inductor would have been, short out the pins of the connections to the HiQ pot and change the Qmax resistor to 4K7.

3 BAND PULTEC STYLE EO



The Pultec 3 band PCB layout is shown below.



Bill Of Materials

The Bill of Materials for the 3 band Pultec EQ PCB is detailed below. As this is an EQ you can modify the component values to alter the EQ frequencies. This BOM only covers the standard frequencies of the the original Poor Man's EQP1A plus the mid frequencies of the Pultec MEQ5.

Switches

The switches listed below fit directly onto the PCB and are also used to attach the complete EQ PCB to the front panel. Alternative lower cost switches can be used if they are connected to the PCB using flying leads.

PCB Ref	Function	Part	Manufacturer
Shi	2 pole 6 way	71BDF30-01-2-AJN	Grayhill
Smid	1 pole 12 way	71BDF30-01-1-AJN	Grayhill
SLo	2 pole 6 way	71BDF30-01-2-AJN	Grayhill

Inductors

Provision is made for two inductors on the PCB; one for high frequency boost and the other for mid frequency boost and cut. The high frequency boost inductor is optional.

PCB Ref	Function	Part	Manufacturer
Lhiboost	1H tapped inductor	VTB9042	Carnhill
Lmid	2H tapped inductor	VTB9050	Carnhill

Potentiometers

The EQ pots are external to the PCB but are included for completeness. The connections for each external pot is brought out to a pre-set pot footprint which is tracked in a standard way. Viewing the footprint as an inverted 'V', the bottom left corner is the fully counter-clockwise end of the pot, the top of the 'V' is the slider and the bottom right corner is the fully clockwise end of the pot.

PCB Ref	Value
HiQ	10K LIN
hiboost	47K/50K LIN
hicut	4K7/5K LIN

Mid	47K/50K LOG
Loboost	47K/50K LOG
Locut	470K/500K LOG
PAD	1K LIN preset

In some countries 4K7 and its multiples are hard to obtain so 5K and its multiples can be used instead. Both values are shown in the table above.

Capacitors

Values of the capacitors required for the mid and lo sections of the EQ are printed directly on the PCB. All capacitors should be metal film types rated at least 63V working. All capacitors need to be radial leaded with a 0.2 inch lead spacing. Suitable 10% tolerance capacitors for all values are available from WIMA. Panasonic also do a nice range of 5% film capacitors for most of the values. The values required and the quantity needed of each are listed in the table below:

Value (nF)	Quantity	WIMA type	Panasonic Type
1	2	FKS2 (5%)	
1.5	1	FKS2 (5%)	
1.8	1	FKS2 (5%)	
2.2	4	FKS2 (5%)	
3.3	3	FKS2 (5%)	
4.7	4	FKS2 (5%)	ECQV (5%)
10	1	MKS2 (10%)	ECQV (5%)
12	2	MKS2 (10%)	ECQV (5%)
18	1	MKS2 (10%)	ECQV (5%)
22	3	MKS2 (10%)	ECQV (5%)
33	1	MKS2 (10%)	ECQV (5%)
47	2	MKS2 (10%)	ECQV (5%)
68	1	MKS2 (10%)	ECQV (5%)
120	1	MKS2 (10%)	ECQV (5%)
150	1	MKS2 (10%)	ECQV (5%)
220	1	MKS2 (10%)	ECQV (5%)
330	2	MKS2 (10%)	ECQV (5%)

For the hi EQ, the capacitor values depend on whether you build the poor man's version or the inductive boost version. The capacitors for the hi boost/cut section are labelled in pairs from C1,C1A to C12,C12A.

PCB Ref	Poor Man's (nF)	Inductive (nF)
C1	47	47
C1A	33	33
C2	47	47
C2A	10	10
C3	47	47
C3A	none	none
C4	22	22
C4A	10	10
C5	22	22
C5A	2.2	2.2
C6	15	15
C6A	none	none
C7	15	4.7
C7A	1	none
C8	10	4.7
C8A	2.2	0.47
C9	10	3.3
C9A	none	none
C10	4.7	1
C10A	1	1
C22	4.7	1
C11A	none	1.5
C12	1	1
C12A	1.5	none

Resistors

All the resistor values are printed on the PCB. All are quarter watt metal film types. The only resistor value that varies is that of Qmax which is located just to the left of the bunch of capacitors below the mid EQ inductor and is labelled 470.

